

ABHISHEK KATOCH

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Python Developer | AI/ML Engineer

PROFESSIONAL SUMMARY

Python Developer with 2+ years of experience in backend development, API design, and scalable system architecture. Strong expertise in building high-performance REST APIs, handling large-scale data processing workflows, and optimizing database queries. Experienced in developing modular and maintainable backend systems using FastAPI and Flask. Currently expanding expertise in Machine Learning and Generative AI, focusing on LLMs, LangChain, and building intelligent AI-powered applications for real-world business use cases.

TECHNICAL SKILLS

Programming & Frameworks — Python, FastAPI, Django, Flask

AI & GenAI — LLMs, LangChain, OpenAI APIs

Data & Libraries — Pandas, NumPy, Matplotlib

Databases — PostgreSQL, MySQL

APIs & Practices — REST APIs, Webhooks, Async Programming, Test-Driven Development

Tools & Practices — Git, Docker, REST APIs, Async Programming

PROFESSIONAL EXPERIENCE

Python Developer — Seasia Infotech, Mohali

- Designed and developed scalable backend systems using FastAPI and Flask, ensuring high performance and maintainability.
- Built robust REST APIs with proper validation, authentication, and structured error handling mechanisms.
- Worked extensively on database schema design, normalization, and query optimization for handling large datasets.
- Implemented asynchronous processing to improve API responsiveness and efficiently handle concurrent user requests.
- Collaborated with frontend developers and product teams to deliver complete end-to-end software solutions.

PROJECTS

Project 1: Task Management System

- Developed a full-featured task management system similar to Jira, supporting complete workflow transitions (To-Do → In Progress → Done).
- Designed and implemented REST APIs for task creation, assignment, updates, and real-time tracking.
- Built scalable and optimized database schema to support multi-user access and high data volume.
- Ensured modular code structure and clean architecture for long-term scalability and maintainability.

Tech Stack: FastAPI, PostgreSQL

Project 2: User Authentication System

- Built a secure authentication system using JWT-based login and signup functionality.
- Implemented role-based access control (RBAC) for managing user permissions and protected routes.
- Applied industry-standard password hashing and token validation techniques.
- Managed token expiration, refresh mechanisms, and middleware-level authentication checks.

Tech Stack: FastAPI, JWT, Security

Project 3:Email Service API

- Developed a reusable email service API for sending transactional and system-generated emails.
- Integrated SMTP-based email delivery with support for dynamic templates.
- Implemented logging, monitoring, and error-handling mechanisms for reliability.
- Designed the service to be easily integrated with multiple backend systems.

Tech Stack: FastAPI, SMTP

Project 4: Notification Service

- Built an asynchronous notification system for handling real-time alerts and updates.
- Implemented retry logic and failure handling to ensure message delivery reliability.
- Used background processing techniques for non-blocking execution.
- Optimized system performance for handling high volumes of notifications.

Tech Stack: Python, Async Processing

Project 5: GenAI Chatbot

- Built an intelligent chatbot using OpenAI APIs and LangChain framework.
- Designed a pipeline for processing user input and generating contextual AI responses.
- Integrated backend APIs for real-time conversational flow.
- Applied prompt engineering techniques to improve response quality and relevance.

Tech Stack: Python, FastAPI, LangChain